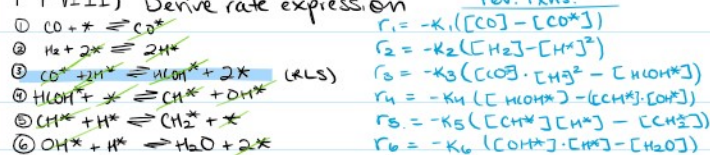


① — Fisher-Tropsch —

FT-VIII) Derive rate expression

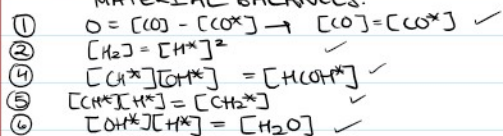


Overall: $\text{CO} + \text{H}_2 + 2\text{H}^* \rightleftharpoons \text{CH}_2^* + \text{H}_2\text{O} + 2*$



$$\frac{r_1}{k_1} = \frac{r_2}{k_2} = \frac{r_4}{k_4} = \frac{r_5}{k_5} = \frac{r_6}{k_6} = 0 \text{ as rxn 3 is RLS.}$$

MATERIAL BALANCES.



$$k_1 P_{\text{CO}} C_v - k_1 ([\text{CO}^*]) = k_2 (P_{\text{H}_2} C_v)^2 - k_2 ([\text{H}^*])^2$$

$$\frac{r_1}{k_1} = 0 = P_{\text{CO}} C_v = [\text{CO}^*] \quad (P_{\text{H}_2} C_v^2 - [\text{H}^*]^2) \frac{k_2}{k_2} = \frac{-r_2}{k_2} = 0$$

$[\text{CO}^*] = P_{\text{CO}} C_v \cdot k_1 = -r_1$

$[\text{H}^*]^2 = P_{\text{H}_2} C_v^2 = 0$

$[\text{CH}_2^*] = \sqrt{P_{\text{H}_2} C_v^2 k_2} = -r_2$

$[\text{CH}^*] = C_v \sqrt{P_{\text{H}_2} k_2}$

$[\text{OH}^*] = C_v k_2$

$k_1 + k_2 = \sqrt{P_{\text{H}_2} k_2}$

Site Balance

$C_T = C_v + [\text{CO}^*] + [\text{H}^*]$
plug in ↓

$C_T = C_v + P_{\text{CO}} C_v \cdot k_1 + C_v k_2$

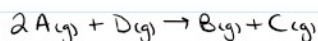
$C_T = C_v (1 + P_{\text{CO}} k_1 + k_2)$

$C_v = \frac{C_T}{(1 + P_{\text{CO}} k_1 + k_2)}$

plug in C_v

$-r_3 = k_3 (P_{\text{CO}}) (C_v) k_1 \cdot C_v \cdot k_2 - \frac{[\text{HCOH}^*] P_{\text{H}_2}^2}{k_3}$

② Rxn occurs on the surface of a catalyst

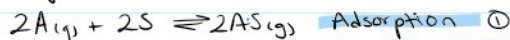


overall rate of rxn ↓

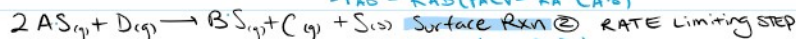
$-r_A = \frac{k_p P_A^2}{(1 + K_A P_A + K_B P_B)^2}$

Suggest a reaction mechanism + show how you would derive above eqn.

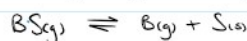
Proposed Mechanism ↓



$-r_{\text{AD}} = k_{\text{AD}} (P_A C_v - \frac{1}{K_A} \text{AS})$



$r_3 = k_3 (C_{\text{AS}}^2 P_D)$



$-r_D = k_D (C_{\text{BS}} - K_B P_B C_v)$

Here, $\frac{r_{\text{AD}}}{k_{\text{AD}}} = \frac{r_D}{k_D} = 0$

Here, $\frac{r_{AD}}{K_{AD}} = \frac{r_D}{K_D} = 0$ *

① $0 = P_A C_V - \frac{1}{K_A} C_A S \downarrow$ ADS.
 $P_A C_V K_A = C_A S \checkmark$

③ $0 = C_B S - K_B P_B C_V \downarrow$
 $P_B C_B C_V = C_B S \checkmark$

Site Balance

$C_T = C_V + C_A S + C_B S$
 $\downarrow$
 $C_T = C_V + K_A P_A C_V + K_B P_B C_V$
 $\downarrow$
 $C_T = C_V (1 + K_A P_A + K_B P_B)$
 $\downarrow$
 $C_V = \frac{C_T}{(1 + K_A P_A + K_B P_B)}$

$-r_D = K_D (C_{B_S} - K_B P_B C_V)$

$r_S = K_S (C_A S)^2 \cdot P_D$

where $C_A S = K_A P_A C_V$

$r_S = K_S K_A^2 P_A^2 C_V^2 \cdot P_D \downarrow$

$r_S = \frac{K_S K_A^2 P_A^2 C_T^2 P_D}{(1 + K_A P_A + K_B P_B)^2}$ $\text{st } K = K_S K_A^2 C_T^2$

$r_S = \frac{K P_A^2 P_D}{(1 + K_A P_A + K_B P_B)^2}$